PROJECT FLOWCHART — ISHAN MEHTA · SEM 2025-26

PHASE 1 · FOUNDATION

Learning & Setup
OSM · Graph Theory · OSMnx · NetworkX · QGIS · GeoPandas

PHASE 2 · DATA PIPELINE

Source Dataset
BBMP PDFs → CSVs (~365 roads)

PREPROCESS

- **Heuristic Filtering**
  Blocklist vague terms · keep anchored locations
- **Build Location Chains**
  Row → [Origin, Via…, Dest]

Geocode
Google Maps API → lat/lng chains

REMOVE REDUNDANCIES

- **Richness Heuristic**
  Same endpoints → keep most via-points
- **Subset Deduplication**
  Remove sub-sequences of longer routes
- 365 → ·273 unique routes

GIS Geometries
Snap to OSM → LineStrings → GeoJSON

API Scraping
Routes API v2 · 84 snapshots / 7 days

PHASE 3 · ANALYSIS (WIP)

PREPROCESS SCRAPED DATA

- **Spatial Deduplication**
  Hash coords → stable segment_id
- **State-Delta Tracking**
  Record only speed-state changes

TRAFFIC ANIMATION (QGIS)

- **Approach 1 — Issues ✗**
  Global dedup → ghost segments
- **Approach 2 — Adopted ✓**
  Per-route · clean & stable

Statistical Analysis
Speed profiles · hotspots · delay distribution

PHASE 4 · LLM PATHFINDING

Context Construction
Roads, times, congestion → LLM input

Query Design
Point-to-point → time-constrained → multi-stop

LLM Candidates
GPT-4 · Claude · Gemini · LLaMA

EVALUATION

- **Ground Truth**
  Google Maps Routes API
- **Metrics**
  Validity · distance · time · overlap

Deliverables
Prompt templates · Benchmark · Failure analysis

⬇ Export HTML 🖨 Print / PDF